SCHOOL OF COMPUTER SCIENCE ENGINEERING | MRCET

(CSE)

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IV Year B.Tech-I Semester Minor Project Individual Summary Sheet

Project Title:	Voice Clone using Deep Learning				
Project Code:	R220A0595	**Batch Size:**	3	**Batch:**	2022-26
Domain / Area:	Artificial Intelligence, Deep Learning, Speech Processing	**SDG Mapping**	9		

Abstract:

This project aims to develop a **Voice Cloning system** using advanced **Deep Learning** techniques integrated with **Eleven Labs API**. Voice cloning is the process of generating synthetic speech that closely mimics a target speaker's voice from a given audio sample and text input. The project uses pre-trained models via Eleven Lab's platform, providing high-quality, human-like voice generation. The main applications include voice assistants, content creation, personalized communication, and accessibility tools.

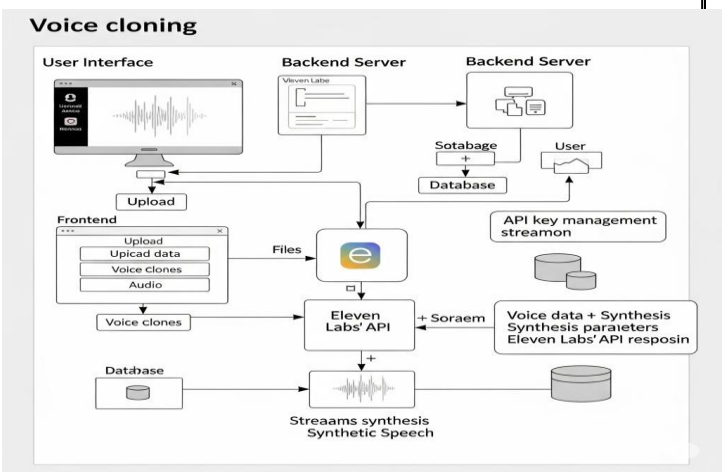
Technical (S/w & H/w) Specifications

- **Software:** Python, Eleven Labs API, Flask (for deployment), Jupyter Notebook.
- **Hardware:** Standard PC with minimum 8GB RAM, internet access.
- **Other Tools:** GitHub, VS Code.

Module(s) Specifications

- Voice Input Capture and Preprocessing.
- API Integration with Eleven Labs.
- Voice Cloning Model Execution.
- Output Synthesis and Playback.
- Web-based.

Architecture Diagram



Methodology

1.Collect voice samples and text input.
2.Use Eleven Lab API to generate cloned voice.
3.Evaluate the audio quality and voice similarity.
4.Provide a simple user interface for testing.
5.Deploy for demo and feedback collection.

Existing System

1. Traditional systems need hours of voice data for training.
2. Voice cloning models are often computationally heavy.
3.Lack of real-time voice synthesis capabilities.
4.Limited accuracy in mimicking emotional tone.

Proposed System

1.Uses deep learning to replicate human voice.
2.Integrated with Eleven Labs API for voice synthesis.
3.Requires minimal voice sample for cloning.
4.Enables applications in AI and accessibility.

Guide Details

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